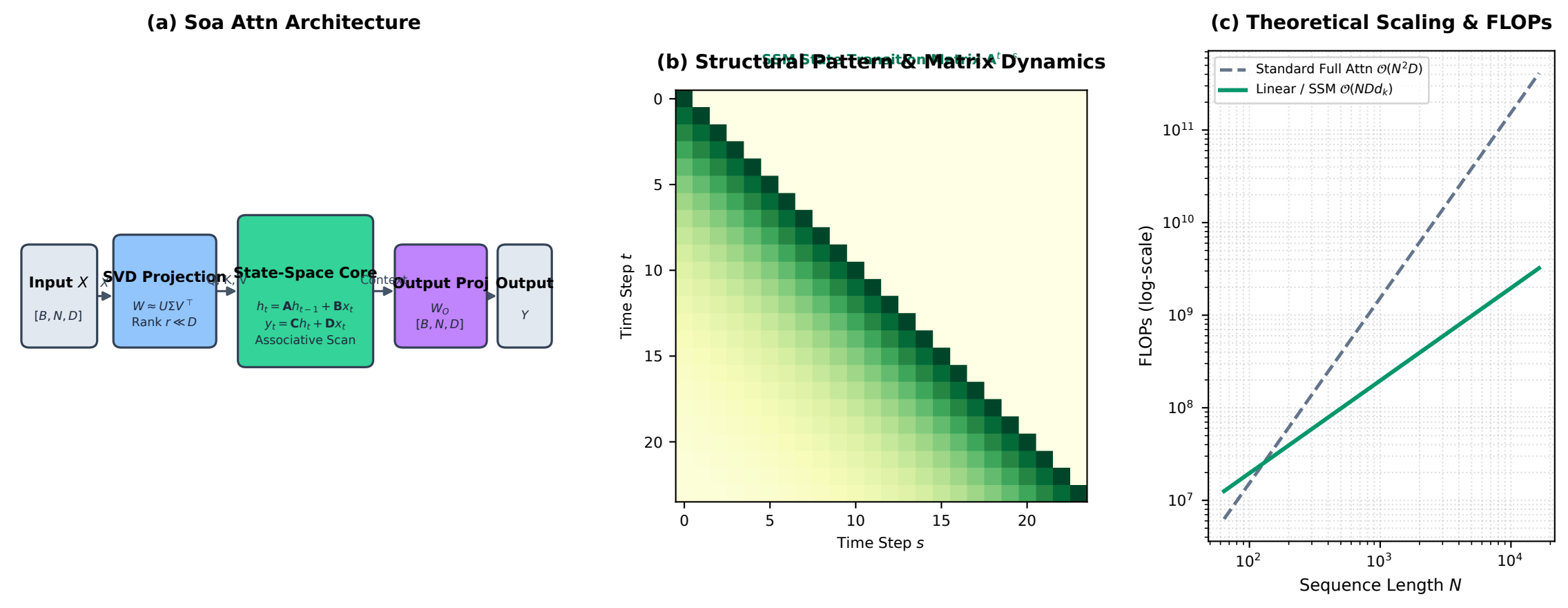
(a) Soa Attn Architecture
Input X
$[B, N, D]$
SVD Projection
$W \approx U\Sigma V^\top$
Rank $r \ll D$
State-Space Core
$h_t = \mathbf{A}h_{t-1} + \mathbf{B}x_t$
$y_t = \mathbf{C}h_t + \mathbf{D}x_t$
Associative Scan
Output Proj
W_O
$[B, N, D]$
Output
Y
X
Q, K, V
Context
(b) Structural Pattern & Matrix Dynamics
SSM State Transition Matrix A^t s
Time Step t
Time Step s
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Linear / SSM $\mathcal{O}(NDd_k)$
FLOPs (log-scale)
Sequence Length N
10^7
10^8
10^9
10^{10}
10^{11}
10^2
10^3
10^4